The Problem in Numbers: U.S. Flight Delays (2003–2022)

Use these figures from the U.S. DOT/BTS "Airline Delay Cause" dataset (covering 121 million flights) in your problem statement and pitch to demonstrate the scale of the issue.

Core Statistics

- **Consistent Delays:** Over the past two decades, approximately 19% (nearly 1 in 5) of all flights arrived 15 or more minutes late. In 2022, this rate rose slightly to 20%.
- **Cancellations:** Roughly 1.9% (about 1 in 50) of all flights are canceled.

Root Causes of Delay

Cause	Share of Delayed Flights	Share of Delay Minutes
Late-arriving aircraft (The Cascade)	34%	38%
Airspace / Airport congestion (NAS)	33%	26%
Carrier (Internal airline issues)	29%	31%
Weather (Direct)	4%	5%
Security	~0%	~0%

Key Takeaways for Modeling

- **The Cascade is the Primary Driver:** Late inbound aircraft constitute the largest single cause of delays. A single delayed flight disrupting subsequent legs is a more frequent problem than direct weather events. This validates the necessity of building the inbound-delay feature using aircraft `tail_number`.
- **Weather is a Catalyst, Not the Whole Picture:** While direct weather delays appear negligible at 4%, weather often triggers airspace congestion and cascade effects. Sample tests indicate that a standalone weather model performs poorly. To be effective, the model must account for the chain reaction rather than isolating weather variables.

Data Source

BTS Airline On-Time Performance, Delay Cause. The complete dataset (segmented by airport, month, and carrier from 2003 to 2022) is available in the build folder for granular analysis.